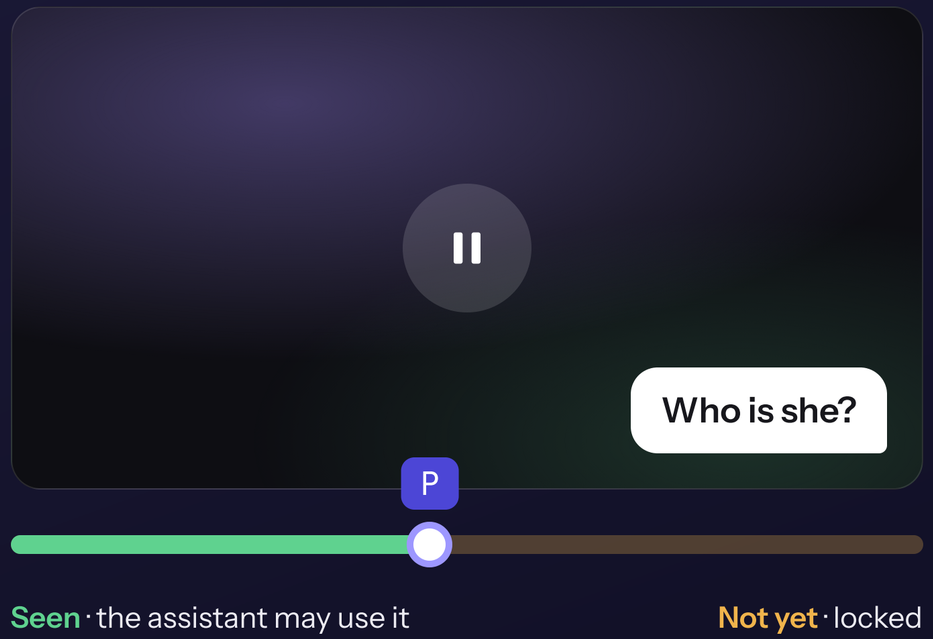
AI BUILDER CUP 2026 · MEDIA, CONTENT & DIGITAL EXPERIENCES

Hello Watch Assistant

A spoiler-free viewing companion, built on Gemini and Google Cloud.

Pause any film and ask. It answers only from what you have already seen: it never names a character before the film does, never hints at a twist and never says an actor's name.



LIVE DEMO hello-watch-vspuwrubbq-uw.a.run.app

CODE github.com/lennonkc/hello-watch-assistant

THE PROBLEM

Viewers pause with questions. Every answer out there is a spoiler risk.

“Who is she?”

A face you can’t place

“What is that?”

An object that seems to matter

“What did I miss?”

You stepped away for ten minutes

“What did he just say?”

A line lost to the soundtrack

A general chatbot knows the whole film

From reviews, wikis and its training data. It cannot tell what you have seen, so it answers with everything it knows.

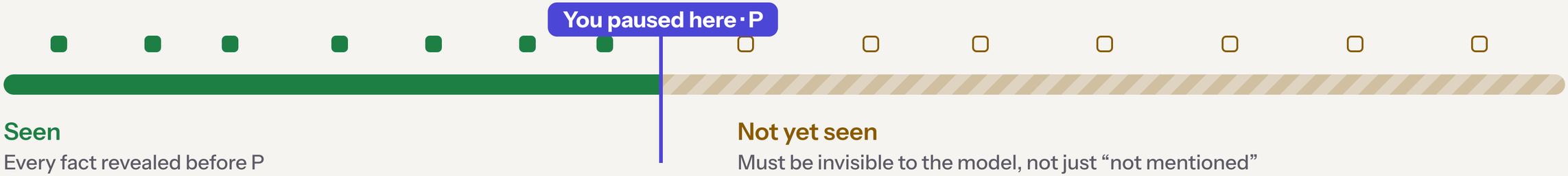
Who is the old man? (minute 12)

He’s the hero’s mentor, who **turns out to be** … — the reveal from minute 80.

Today viewers rewind, search the web and risk spoilers, or stay confused.

WHY IT IS HARD

Spoiler safety can’t be a polite instruction to the model.



Models remember plots

Popular films are in the training data. Telling the model “don’t spoil” leaves the knowledge in place.

Search ranks the future too

Retrieval over a whole film puts later scenes in the candidate list. One missed filter and they reach the prompt.

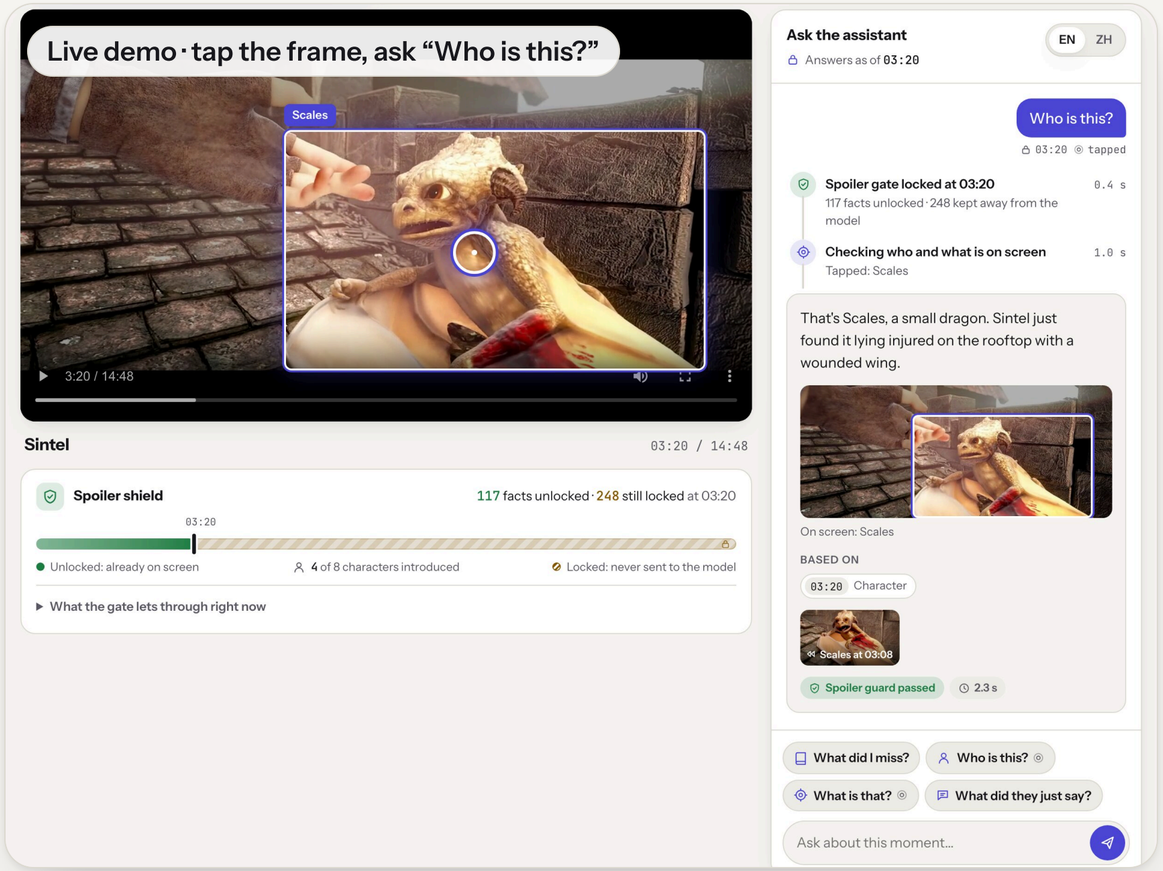
Leaks look like good answers

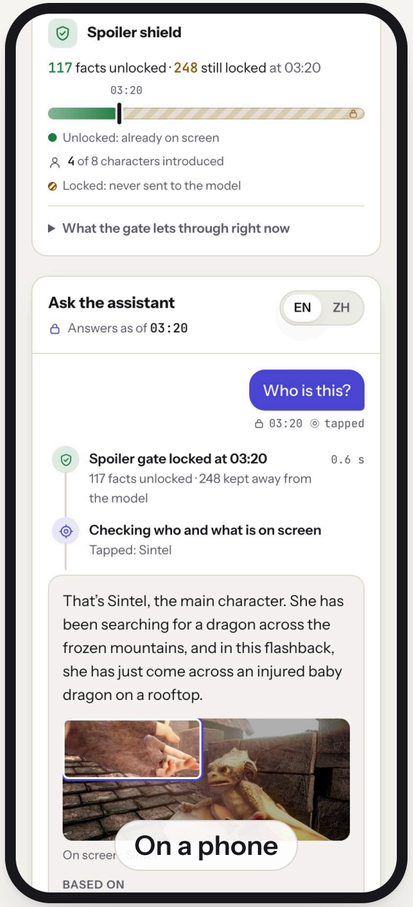
A spoiled answer reads fluently and confidently. Nothing in the output signals that something went wrong.

So the model must never see what the viewer hasn’t seen. The gate has to be structural.

THE EXPERIENCE

Pause, tap, ask. The answer stops exactly where you are.





Four everyday questions

Who someone is, what an object is, the story so far, and the line you just missed.

Tap the frame

The tap picks the person or object; the answer draws a box around them.

See how it answered

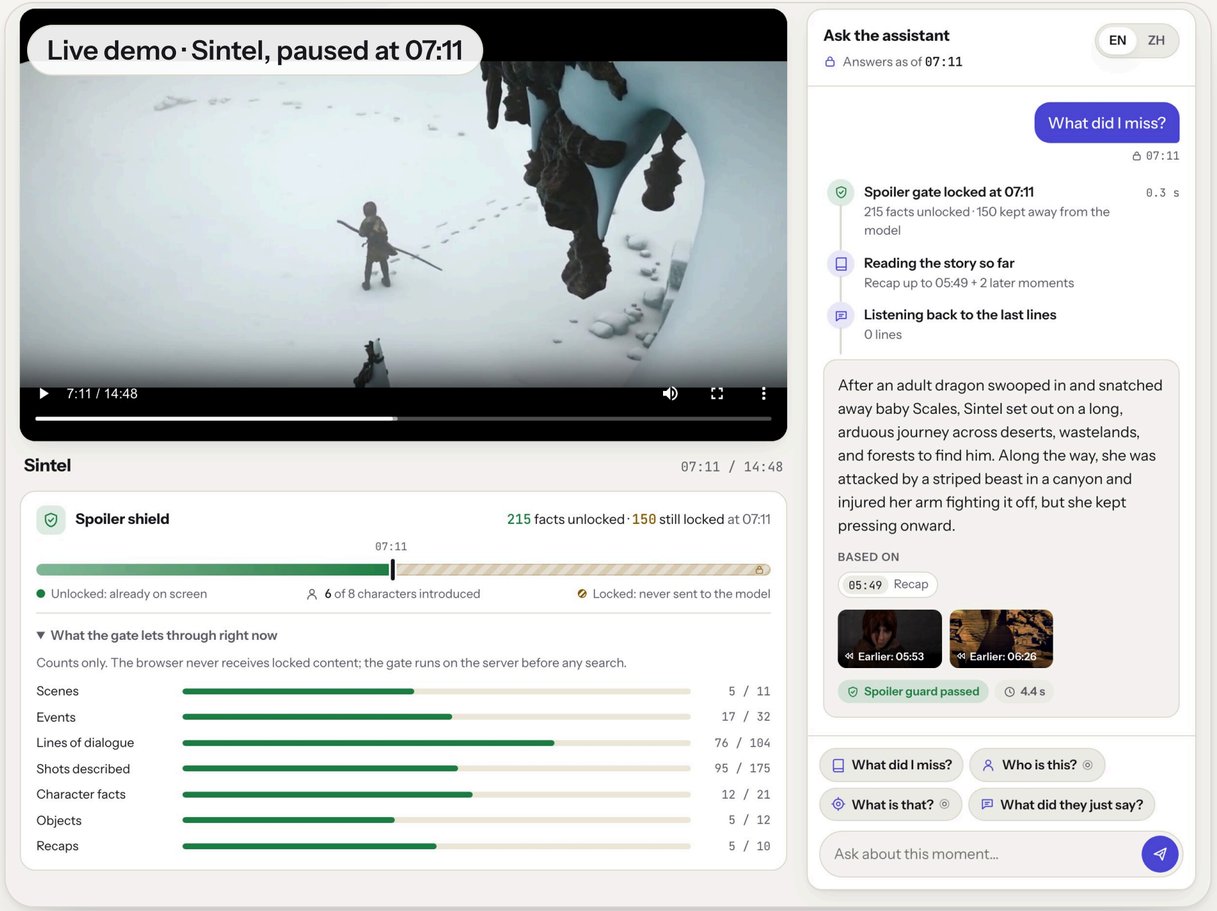
Each step streams in live, with evidence frames all from before P.

English and Chinese

Ask and get answers in either language.

THE EXPERIENCE

“What did I miss?” recaps the story up to your exact moment.



The gate runs first

At 07:11, 215 facts are unlocked and 150 are kept away from the model.

Story so far

The latest recap plus what happened since, read by the agent’s story_so_far tool.

Evidence you can check

“Based on” shows the recap time and frames from earlier in the film.

Guarded and fast

The spoiler guard passed; this answer took 4.4 s.

HOW IT WORKS

Gemini studies each film once. Every question then passes the gate first.

A · Offline ingest once per film · local or Cloud Run Jobs

Source film
Cloud Storage

→

Shots and frames
FFmpeg · PySceneDetect

→

Video understanding
Gemini 3.8 Flash: characters, shots, speakers, recaps; gemini-3.5-transcribe for dialogue

→

Embeddings
gemini-embedding-2, text and video

→

Builder checks
11 automated checks

→

Knowledge pack
SQLite, a reveal time on every row

The pack stays on the server. Only the gate reads it.

B · Online Q&A every question · Cloud Run service

Pause and ask
Question, position, optional tap

→

Lock P
On the server, in session state

→

Spoiler gate
Eligible set: reveal + margin ≤ P

→

ADK agent + 5 tools
Gemini 3.8 Flash on Agent Platform; tools read only the eligible set

→

Output guards
Names, canaries, visibility

→

Streamed answer
Server-Sent Events

THE SPOILER GATE

Gate first, then rank. The model can’t leak what it was never shown.

How eligibility works

- ✓ **P is locked on the server.** The browser reports a position; it never sends, or receives, the list of what is visible.
- ✓ **Every row has a reveal time.** It becomes eligible when `reveal_at_s + margin ≤ P`, with margins per type: dialogue 1 s, events 3 s, scenes 3–5 s.
- ✓ **Names wait for the film.** A character is “the young woman” until the film says her name; only then may the name be used.
- ✓ **Identities stay apart until a reveal.** Facts are never linked across a hidden identity; the clue to it is a future canary.
- ✓ **Ranking runs inside the gate.** Vector, full-text or hybrid ranking only ever sees the eligible set.

Rejected on purpose

- ✗ **Managed RAG with metadata filters**
  Ranks the whole film, then filters. If the filter silently stops applying, the result is a spoiler.
- ✗ **Grounding with online search**
  Brings in synopses, reviews and wikis: exactly where spoilers live.
- ✗ **Calling Gemini from the browser**
  The client would need the evidence, and a client can be modified. The gate must run behind the server’s trust boundary.

THE AGENT

An ADK agent with five gate-aware tools, and no way to move P.

`story_so_far`	Latest eligible recap plus the scenes and events after it. The first choice for “what did I miss?”
`search_moments`	Embeds the question with gemini-embedding-2 and ranks only inside the eligible set.
`recent_dialogue`	Lines spoken before P, with who said each one when known.
`on_screen`	Who and what is in the paused frame; a tap is matched to their boxes, and labels are decided at P.
`look_closer`	Gemini 3.8 Flash looks at the paused frame plus stills from the 30 s before it.

P lives in session state

The server writes P, title, language and tap into the ADK session with `state_delta`. Tools read `ToolContext.state`; no tool takes P as an argument.

Every step is visible

`gate` → `tool` → `tool_result` → `answer`

Streamed to the browser with Server-Sent Events.

Guard, retry, fall back

If the output guard finds a forbidden term, the draft is discarded and the question is answered again in a fresh session with a do-not-use hint. If that still fails, a fixed safe answer is returned.

BUILT ON GOOGLE CLOUD

Twelve Google services, each with one clear job.

GEMINI

Gemini 3.8 Flash

Watches each film during ingest; powers the agent and visual look-ups online.

GEMINI

gemini-embedding-2

One 1536-d space for text items and ≤ 30 s video windows with audio.

GEMINI

gemini-3.5-transcribe

Word-level transcription with speaker diarization.

AGENTS

Gemini Enterprise Agent Platform

Serves Gemini with project IAM; reads video straight from Cloud Storage.

AGENTS

Agent Development Kit

The online LlmAgent and its five gate-aware tools.

COMPUTE

Cloud Run service

Web app, spoiler gate, agent and output guards; capped instances and rate limits.

COMPUTE

Cloud Run Jobs

Batch ingest: one film per task, cached and budgeted stages.

DATA

Cloud Storage

Films and versioned knowledge packs in a private bucket; video via signed URLs.

SECURITY

Secret Manager

Every production prompt, the agent instructions and the API key. None are in the repo.

BUILD

Cloud Build

Builds the ingest image (pinned FFmpeg, Python) and the web service image.

BUILD

Artifact Registry

Stores the images that Cloud Run runs.

OPERATIONS

Cloud Scheduler

Triggers the upload-renewal job every 6 hours.

RESULTS

Measured on three open films, processed end to end.

0 / 120
gate violations in checked retrieval results

36 / 40
vector retrieval hits in the top 3 (full-text only: 22 / 40)

12 / 14
Chinese questions with a correct top-3 retrieval hit (full-text: 0 / 14)

10 / 10
speaker attributions correct in spot checks

12 / 12
character identities correct in shot entities

~85%
shot captions fully correct in spot checks

3–13 s
to a full answer (local tests), with ≤ 3 tool calls by instruction

≈ $8
total Gemini cost to ingest all three films, re-runs included

Spot checks on Sintel unless noted. Retrieval: 20 questions × 2 pause positions, scored by top-3 time overlap. Films: Sintel, Tears of Steel, Elephants Dream (Blender Foundation, CC BY).

COMMUNITY BENEFIT

Help that respects the story, for people who need it most.

ACCESSIBILITY

Deaf and hard-of-hearing viewers

“Who said that?” is answered from speaker-attributed dialogue, without spoiling what comes next.

LANGUAGE

Viewers watching in a second language

Ask in Chinese about an English film. Cross-language retrieval finds the moment, and the answer comes back in your language.

ATTENTION

Viewers who step away

A recap that stops exactly where you are, for families, interrupted evenings and long films.

Education and libraries

A study companion for film classes and library collections that explains what students have watched without giving away the ending.

Streaming, TV and creators

A second-screen companion that keeps viewers engaged instead of searching elsewhere. Creators can open their story to questions because reveal times are data they can review.

FEASIBILITY AND SCALABILITY

Built, deployed and cheap to run, with a clear path to scale.

Feasible today

- **Live on Cloud Run** with three films, a public link and an open-source codebase.
- **Ingest once per film.** Cached, budgeted stages; about $8 of Gemini for all three films, re-runs included.
- **Byte-reproducible packs:** identical content re-exports to identical bytes and the same version.
- **Guarded spend:** at most 2 instances, per-IP and daily question limits.

Ready to scale

- **Cloud Run Jobs** ingest one film per task, in parallel.
- **The knowledge pack is versioned SQLite.** A device form (reviewed rows only, no review log) is already defined, designed for answering on a TV or phone.
- **Prompts in Secret Manager** change without rebuilding images.
- **Human review is specified:** review patches, roles and a rule that names, identities and spoiler-critical items need a human.

Honest limits: the demo serves automatically pre-checked rows; production would add the human review step for names and twists. Shot captions are about 85% fully correct, and the output guards are a safety net, not a guarantee.

ROADMAP

From three open films to any catalogue, on any screen.

NEXT

Voice questions

Push-to-talk or a wake word. P locks at the moment you speak, as the spec already defines.

NEXT

TV and mobile apps

Ship the device pack with the title so simple questions work offline on the living-room screen.

NEXT

Reviewer tooling

A review interface on top of the existing patch format, so a person can confirm names and reveals quickly.

LATER

Series memory

Gate across episodes and seasons: “previously on” recaps that stop at your episode.

LATER

More languages

Recaps and answers beyond English and Chinese, with name canaries for each language.

LATER

Partner catalogues

Studios, broadcasters and libraries ingest their own titles with their own review.

HELLO WATCH ASSISTANT

Ask about the film. Never get spoiled.

LIVE DEMO https://hello-watch-vspuwrubbq-uw.a.run.app

CODE https://github.com/lennonkc/hello-watch-assistant

RESEARCH IT BUILDS ON

AutoAD II and **AutoAD-Zero**: in-film character bank, circle visual prompting

Shot-by-Shot: shot-level description

MovieChat: breakpoint mode, progressive recaps

HAVEN and speaker-aware long-video QA

DEMO FILMS

Sintel © Blender Foundation, durian.blender.org, CC BY 3.0

Tears of Steel © Blender Foundation, mango.blender.org, CC BY 3.0

Elephants Dream © 2006 Blender Foundation / Netherlands Media Art Institute, orange.blender.org, CC BY 2.5